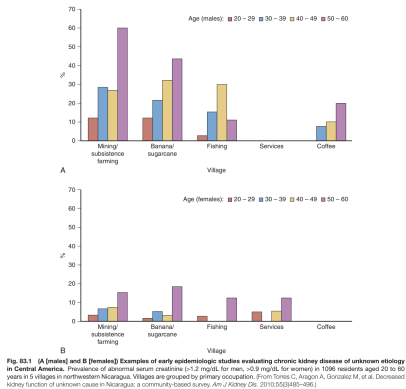




Fig_83_1

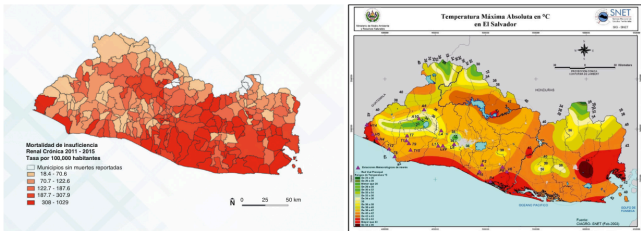


Table 83.1 Similarities and Differences in Clinical Presentation and Other Associated Features of Chronic Kidney Disease of Unknown Etiology in the 2 Best-Described Hotspots			
	Mesoamerica	Sri Lanka	
Geospatial Factors			
Hot, dry climate	✓	✓	✓
Reliance on groundwater supply	✓	✓	✓
High-density agrochemical application	✓	✓	✓
Low population-based prevalence of diabetes	✓	✓	✗
Clinical Presentation/Other Associated Features			
Male predominance	✓	✓	✓
Lack of heavy proteinuria	✓	✓	✗
Significant hypercemia	✓	✓	✗
Significant hypokalemia	?	?	?
Extensive use of NSAIDs	✓	✓	✗
Specific job tasks with higher risk	?	?	✗
Common tobacco use	?	?	✓
Biopsy Data			
Chronic tubulointerstitial nephritis and fibrosis	✓	✓	✓
Some with primarily acute interstitial nephritis	✓	✓	✓
No immune complex deposition	✓	✓	✓
Significant glomerulosclerosis	✓	✓	?
Question mark symbols indicate lack of definitive data to make a determination. NSAIDs, Nonsteroidal antiinflammatory drugs.			

Table_83_1

Table 83.2 Summary of Chronic Kidney Disease of Unknown Etiology Detection Efforts in Cross-Sectional Population-Based Studies												
Reference	Year Published	Sub-Continent	Country and Region	N	Exclusion of Diabetes	Exclusion of Hypertension (Threshold)	Exclusion of Proteinuria (Threshold)	Other Exclusions	Chronicity Demonstrated	% eGFR <60 mL/min/1.73 m ²	% eGFR <60 mL/min/1.73 m ² as defined by eGFR <60 mL/min/1.73 m ²	Independent Risk Factors for CKD/CKDu as defined by eGFR <60 mL/min/1.73 m ²
O'Donnell et al. ¹⁰	2011	Central America	Leon, Nicaragua	771	N	N	N		N	13%		Age Male sex Low altitude Age Male sex Agricultural or mining communities
Torres et al. ¹⁰	2010	Central America	Leon and Chinandega, Nicaragua	1096	N but low prevalence	N	N		N	8.7%		Age Male sex Agricultural or mining communities
Perez et al. ¹⁰ https://doi.org/10.1053/ajkd.2011.11.039	2012	Central America	Pacific Coast, El Salvador	664	N but low prevalence	N	N		N	0% in high altitude/urban areas 10% in sugarcane communities		Age Male sex Low Altitude Sugarcane/cotton work Tobacco use (males only) Kidney stones (males only) Diabetes (females only) Hypertension (females only)
Orantes et al. ¹²	2011	Central America	Bajo Lempa, El Salvador	775	Y	Y	N		Y	3.6%		Age Male sex Family history of CKD
Milar et al. ¹⁶ https://doi.org/10.1016/j.kid.2020.12.015	2021	Central America	Tecpan, Guatemala and San Antonio Suchitupé, Guatemala	807	N	N	N		N (Sensitivity 4.0% analysis only)			Age Diabetes Hypertension Low BMI Sugarcane work in poorer participants

Continued on following page

Table_83_2_part1

Table 83.2 Summary of Chronic Kidney Disease of Unknown Etiology Detection Efforts in Cross-Sectional Population-Based Studies (Continued)												
Reference	Year Published	Sub-Continent	Country and Region	N	Exclusion of Diabetes	Exclusion of Hypertension (Threshold)	Exclusion of Proteinuria (Threshold)	Other Exclusions	Chronicity Demonstrated	% eGFR <60 mL/min/1.73 m ²	% eGFR <60 mL/min/1.73 m ² as defined by eGFR <60 mL/min/1.73 m ²	Independent Risk Factors for CKD/CKDu as defined by eGFR <60 mL/min/1.73 m ²
Ruiz-Alajos et al. ¹⁰	2021	South America	Tumbes, Peru	1,514	Y	Y (140/90)	Y		N	5.9%		Numbers too small to undertake analyses in those without hypertension or diabetes
O'Callaghan-Gordo et al. ¹⁰	2019	South Asia	Chennai, Visakhapatnam, Hyderabad and Delhi, India	12,500	Y	Y (140/90)	Y		N	1.4% Urban 1.9% Rural	1.4% Urban 1.9% Rural	Age Northern areas/Rural residence (Southern areas only) Southern areas only Fewer years of education
Tatapudi et al. ¹⁶	2019	South Asia	Udumaluru, India	2210	Y	Y (140/90 for more than 5 years)	Y (PCR >100 mg/g)		N	15.3%		Age
Gummididi et al. ¹⁷	2020	South Asia	Udumaluru, India	2419	N	N	N		Y	10.2%		Age Male sex Hypertension Tobacco use Lack of formal education Outdoor work Family history of CKD
Ruvangathirana et al. ¹²	2019	South Asia	Anuradhapura district, Sri Lanka	4803	Y	Y (140/90)	Y (ACR >30 mg/g)			3.7% in lowest impacted area 8.5% in highest impacted area	3.7% in lowest impacted area 8.5% in highest impacted area	Analgesic use Age Male sex Family history of CKD Tobacco use (males only)

Table_83_2_part2

Table 83.2 Summary of Chronic Kidney Disease of Unknown Etiology Detection Efforts in Cross-Sectional Population-Based Studies (Continued)												
Reference	Year Published	Sub-Continent	Country and Region	N	Exclusion of Diabetes	Exclusion of Hypertension (Threshold)	Exclusion of Proteinuria (Threshold)	Other Exclusions	Chronicity Demonstrated	% eGFR <60 mL/min/1.73 m ²	% eGFR <60 mL/min/1.73 m ² as defined by eGFR <60 mL/min/1.73 m ²	Independent Risk Factors for CKD/CKDu as defined by eGFR <60 mL/min/1.73 m ²
Jayatilake et al. ¹⁰	2013	South Asia	Anuradhapura, Badulla, Hambantota and Polonnaruwa districts, Sri Lanka	4957	Y	Y (160/90)	N		N	NA	NA	Proteinuria screening only
Hathaway et al. ¹⁰	2023	Sub-Saharan Africa	Kisumu, Kenya	782	Y	Y	Y	HIV	N	0%	NA	
Hamilton et al. ¹⁰	2020	Sub-Saharan Africa	Lilongwe and Karonga district Malawi	821	Y	Y (140/90)	Y			0.2%	NA	
Aekplakorn et al. ¹⁴	2023	Southeast Asia	National, Thailand	17,329	Y	Y (140/90)	Y (urinalysis >1+)	Age>70	N	0.7% Central, South and Bangkok regions 1.6% Northeast region	0.7% Central, South and Bangkok regions 1.6% Northeast region	Age Female sex Gout Kidney stones Rural residence Analgesic use Manual labour/agricultural work Low BMI

Table_83_2_part3

Table 83.3 Hypothesized Potential Cause(s) of Chronic Kidney Disease of Unknown Etiology (CKDu)	
Proposed Causes of CKDu	Example Supportive Evidence
Occupational heat exposure	Cross-shift change in serum creatinine ^{10,15} Cross-harvest change in eGFR among those with most strenuous tasks ^{10,12}
Metals and metalloids	Fluoride ¹¹ High overall mineral content ("hard water") ^{10,11} Nickel ¹¹ Silica nanoparticles ¹⁴ Other metals including arsenic
Agrochemicals	Glyphosate ^{14,16} Paraquat ¹⁷ Mixture of agrochemicals through well water ^{10,16}
Infections	Leptospirosis ¹⁸ Hantavirus ¹⁶ Cyanobacteria ¹⁸
Genetic	Proximal tubular potassium channel polymorphism ¹⁰

Table_83_3